

HTCS6707 Assignment

Part 1: Project Intent Proposal

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Dataset Identification

Dataset: Phishing Website Dataset

Source: <https://archive.ics.uci.edu/dataset/327/phishing+websites>

Description:

The dataset I chose for this project is the Phishing Website Dataset from UCI Machine Learning Repository, created by Rami Mohammad and Lee McCluskey. The dataset file is a collection of features extracted from legitimate and phishing websites to help us distinguish between a legitimate website and a fraudulent one. It features total of 31 attributes represented by -1 (Phishing), 0 (Suspicious), or 1 (Legitimate):

Attribute	Possible values	Description
1. having_IP_Address	-1: Website uses an IP address in URL 1: Otherwise	If the domain part has an ip address (e.g. http:192.168.1.10), it is a phishing website, otherwise legitimate.
2. URL_Length	-1: URL length is longer than 75 0: URL length is between 54 and 75 1: URL length is shorter than 54	Phishers use long URL to hide the doubtful part in the address bar. In this dataset, if the length of the URL is greater than or equal to 54 characters then the URL can be suspicious of phishing website.
3. Shortining_Service	-1: URL uses shortening services	URL may be made considerably smaller in length and still lead to the required webpage.

	1: Otherwise	If the URL uses URL shortening such as 'bit.ly', it can be considered as phishing.
4. having_At_Symbol	-1: URL has @ symbol 1: Otherwise	Using "@" symbol in the URL leads the browser to ignore everything preceding the "@" symbol and the real address often follows the "@" symbol.
5. double_slash_redirecting	-1: Position of '/' appears after seventh position 1: Otherwise	The existence of "/" within the URL path means that the user will be redirected to another website. In the URLs starting with 'http:', '/' appears in the sixth position and in the URLs starting with 'https:', '/' appears in the seventh position.
6. Prefix_Suffix	-1: Domain name part includes '-' 1: Otherwise	Dash is rarely used in legitimate URL. Phishers tend to add prefixes or suffixes separated by (-) to the domain name so that users feel that they are dealing with a legitimate webpage.
7. having_sub_Domain	-1: Otherwise 0: Two dots in the URL 1: One or no dot in the URL	From the URL, after removing 'www.' and country-code top-level domains (ccTLD) e.g. '.uk', count the number of '.'. If the number of dots is greater than one, then the URL is classified as "Suspicious" since it has one sub domain. However, if the dots are greater than two, it is classified as "Phishing" since it will have multiple sub domains. Otherwise, if the URL has no sub domains, we will assign "Legitimate" to the feature.
8. SSLfinal_State	-1: Otherwise	It is suggested that we check the certificate assigned with HTTPS

	0: Using certificate issuer from non-trusted 1: Using certificate issuer from trusted and age of this certificate is greater or equal two one year	including the extent of the trust certificate issuer, and the certificate age. top trustworthy names include: “GeoTrust, GoDaddy, Network Solutions, Thawte, Comodo, Doster and VeriSign”. In this dataset, the minimum age of a reputable certificate is two years.
9. Domain_registration_length	-1: Domain expires on one year or below 1: Otherwise	Phishing websites live for a short period of time and legitimate domains are regularly paid for several years in advance. In this dataset, the longest fraudulent domains have been used for one year only.
10. Favicon	-1: Favicon loaded from external domain 1: Otherwise	Favicon is a graphic icon associated with a specific webpage. Many existing user agents show favicon as a visual reminder of the website identity in the address bar. If the favicon is loaded from a domain other than that shown in the address bar, then the webpage is likely to be considered as a phishing attempt.
11. port	-1: Important port(s) is / are not in the preferred state 1: Otherwise	To control intrusion, it is recommended to only open the ports that we need and block the rest. If all the ports are opened, attackers can run almost any services they want. The most important ports and their preferred status are: • 21 (FTP) - Close • 22 (SSH) - Close

		• 23 (Telnet) - Close • 80 (HTTP) - Open • 443 (HTTPS) - Open • 445 (SMB) - Close • 1433 (MSSQL) - Close • 1521 (ORACLE) - Close • 3306 (MySQL) - Close • 3389 (Remote Desktop) - Close
12. HTTPS_token	-1: 'https' is part of the URL's domain 1: Otherwise	Phishers may add 'https' to the domain part of a URL to trick users. (e.g. http://https-www....)
13. Request_URL	-1: More than 22% Request URL are from another domain 1: 22% or below Request URL are from another domain	Request URL examines whether the external objects contained within a webpage such as images, videos and sounds are loaded from another domain. In legitimate webpages, the webpage address and most of objects embedded within the webpage are sharing the same domain.
14. URL_of_Anchor	-1: Otherwise 0: Between 31% ~ 67% of anchor links to different domain or webpage 1: Below 31% of anchor links to different domain or webpage	An anchor is an element defined by the tag <a>, a 'request URL'. If too many <a> tags link to any website with different domain name or link to any webpage (e.g. , , it is considered as phishing website.
15. Links_in_tags	-1: Otherwise	Legitimate websites use: • <Meta> tags to offer metadata (a data that

	0: Between 17% to 81% of <Meta>, <Script> and <Link> tags are not linked to the same domain of the webpage 1: Below 17% of <Meta>, <Script> and <Link> tags are not linked to the same domain of the webpage	describes other data) about the HTML document. • <Script> tags to create a client-side script • <Link> tags to retrieve other web resources If these tags are not linked to the same domain of the webpage, they can be classified as suspicious.
16. SFH	-1: SFH is empty or “about:blank” 0: SFH refers to a different domain 1: Otherwise	Server Form Handler (SFH) that contain an empty string or “about:blank” are considered suspicious because an action should be taken upon the submitted information. And if the domain name in SFHs is different from the one from the webpage, it is suspicious because the submitted information is rarely handled by external domains.
17. Submitting_to_email	-1: Uses mail() or mailto function to submit user information 1: Otherwise	Web form allows a user to submit his personal information that is directed to a server for processing. A phisher might redirect the user’s information to his personal email by using a server-side script language such as “mail()” function in PHP or a client-side function such as “mailto” function.
18. Abnormal_URL	-1: The hostname is not included in URL 1: Otherwise	In legitimate websites, their identities are part of their URL.

19. Redirect	0: Website has been redirected between 2 ~ 3 times 1: Website has been redirected one time max	How many times has a website been redirected. In this dataset, legitimate websites have been redirected one time max. If the websites have been redirected at least 4 times, they can be classified as phishing websites.
20. on_mouseover	-1: onMouseOver changes status bar 1: It doesn't change status bar	Phishers may use JavaScript to show a fake URL in the status bar to users. We can check 'onMouseOver' event in the webpage's source code and check if it makes any changes on the status bar.
21. RightClick	-1: Right click disabled 1: Otherwise	Phishers often use JavaScript to disable the right-click function, so taht users cannot view and save the webpage source code.
22. popUpWidnow	-1: Popup window contains text fields 1: Otherwise	It is unusual for legitimate websites to ask users to submit personal information through a popup window.
23. Iframe	-1: Using iframe 1: Otherwise	Phishers use iframe tag and make additional webpage invisible (without frame borders) which causes the browser to render a visual delineation.
24. age_of_domain	-1: Otherwise 1: Age of domain is 6 months or longer	Most phishing websites live for a short period of time. In this dataset, the minimum age of the legitimate domain is 6 months.
25. DNSRecord	-1: No DNS record for the domain	Phishing websites do not have records of hostnames.

	1: Legitimate	
26. web_traffic	-1: Domain has no traffic or is not recognised by the Alexa database -0: Website rank is below top 100,000 1: Website rank is within top 100,000	The popularity of the website. In this dataset, legitimate websites are ranked among the top 100,000. Because phishing websites live for a short period of time, they may not be recognised by the Alexa database.
27. Page_Rank	-1: PageRank is below 0.2 1: Otherwise	PageRank is a value ranging from 0 to 1 and measures how important a webpage is on the internet. Greater the PageRank value, the more important the webpage. In this dataset, about 95% of phishing webpages have no PageRank and the remaining 5% only have PageRank up to 0.2.
28. Google_Index	-1: Otherwise 1: Webpage Indexed by Google	This attribute examines whether a website is in Google's index or not. If a site is indexed by Google, it is displayed on search results.
29. Links_pointing_to_page	-1: Number of links pointing to the webpage is 0 0: Number of links pointing to the webpage is between 1 ~ 2 1: Otherwise	The number of links to the webpage indicates its legitimacy level. In this dataset, 98% of phishing websites have no links pointing to them and legitimate websites have at least 2 external links pointed to them.

30. Statistical_report	-1: Host belongs to top phishing IPs or Top phishing domains 1: Otherwise	In this dataset, authors used phishing statistical reports of top 10 domains and top 10 IPs from 'PhishTank' between 2010 ~ 2012 and top 50 IPs from 'StopBadware' for their research.
31. Result	-1: Phishing 1: Legitimate	Determines whether the website is phishing or legitimate based on combination of other 30 attributes.

Cybersecurity Questions

Three questions should reflect on the following three data analysis techniques:

- **Descriptive data analysis**
- **Exploratory data analysis**
- **Predictive data analysis**

Question 1: What are the most common features of phishing websites in the dataset?

This question reflects descriptive data analysis. It aims to summarise statistical data of the main features of phishing websites to help understand the general profile of phishing websites compared to legitimate ones.

Question 2: Are there any significant patterns or correlations among the feature set that help distinguish phishing sites from legitimate ones?

This question reflects exploratory data analysis. It aims to discover potential patterns or relationships between attributes and identify which features might be the important indicators from visualisation methods such as histogram, pie chart, box plots, or scatter plots.

Question 3: Can we build a machine learning model to predict phishing websites?

This question reflects predictive data analysis. It aims to build predictive models from the dataset and measure its performance on how well it can detect phishing.

Preliminary Analysis Plan

This analysis plan focuses on performing a comprehensive data analysis including exploring relationships between features, identifying common phishing patterns, building a predictive model, and interpreting the results to provide meaningful insights into identifying phishing websites.

Key Steps:

1. Data preparation

- a. Import the dataset file, training.arff on Jupyter notebook.
- b. Process data cleaning tasks: inspect any missing values or data inconsistencies and clean the data as needed.

2. Descriptive Data Analysis

- a. Generate a statistical summary of mean, median, mode, and frequency for all attributes.
- b. Use frequency counts of phishing and legitimate URLs using the 'Result' attribute to show class balance.

3. Exploratory Data Analysis

- a. Using visualisation methods such as histogram, scatter plot, correlation heatmap, histogram, pie chart or box plot (to compare distribution), find attributes most associated with phishing URLs.
- b. Identify which attributes have strong separability between phishing and legitimate websites.

4. Predictive Data Analysis

- a. Build classification models such as regression, time series model, or machine learning to predict phishing outcomes using the dataset features.
- b. Evaluate models' accuracy, precision, ROC (Receiver Operating Characteristic) and AUC (Area Under the Curve).
- c. Compare models and select the model with the best phishing detection.

5. Conclusion

- a. Summarise the findings from the three data analyses and discuss their impact on cybersecurity practices in a report.

Project Timeline

Week	Tasks
5	Part 1: Read the assignment, choose dataset and plan
6	Part 1: Write Project Intent Proposal
7	Part 1: Fix any errors and submit Project Intent Proposal
8	Part 2: Use Jupyter Notebook or Google Colab to clean data from the chosen dataset, conduct analysis and evaluate
9	Part 2: Collect all the data and statistics from chosen dataset and write introduction for the report
Holiday week 1	Finish off any remaining work
Holiday week 2	Finish off any remaining work
10	Part 2: Write descriptive and exploratory analysis in the report
11	Part 2: Apply predictive analytics and write them in the report
12	Part 2: Write conclusion, proof-read and fix any errors
13	Part 3: Summarise the dataset, questions, process, findings, and recommendations in one Powerpoint file
14	Part 3: Record a 10-minute narration with powerpoint
15	Submit part 2 & 3